TEAM LORA FARM

QUALCOMM AI INCIDENT REPORT ASSISTANT

- *ABSTRACT*
- **Incident reporting can be time-consuming when details are recorded manually, often resulting in incomplete or inconsistent reports. The Qualcomm AI Incident Report Assistant, developed for Megathon X at IIIT Hyderabad, aims to simplify this process by converting natural conversational speech into a structured incident report using on-device generative AI.**

- **The proposed workflow captures speech through a microphone, transcribes it using speech-to-text (STT), and extracts relevant information with an on-device language model. Extracted details may include incident type, severity, location, date and time, people involved, assets, incident description, immediate actions, witness information, and recommended follow-up actions. The application validates the extracted data, preserves unknown values rather than inventing details, and checks whether mandatory fields are complete. If information is missing, it asks a focused follow-up question and merges the response into the existing report.**

- **Users can review, correct, and confirm the generated report before finalization. The solution is intended to run entirely on the Qualcomm HDK8750 (SM8750P) development platform with Snapdragon acceleration, without cloud APIs or network-dependent inference. On-screen output and text-to-speech (TTS) are planned to support accessible interaction. The core objective is a reliable, privacy-conscious incident-reporting workflow that can operate offline on the target device.**

- ***Keywords:*** **Incident reporting, conversational speech, speech-to-text, on-device generative AI, structured information extraction, offline processing.**